11 Deterministicbuild

Lois-Kleinner Alpasan · Anticloud FZ LLE · PAX L5 Narrow L2 General 27B

Deterministic Build & Single Binary — WRF$_{PYTHON}$

Company: Anticloud FZ LLE

1. Build Metrics

Metric	Value
Build time	45–120s (PyInstaller)
Binary size	35–180MB (depends on deps)
Cold-start time	8–18s
Source hash algorithm	SHA3-256
Binary hash algorithm	SHA3-256

2. Determinism Test

Same source → same binary hash across:

- Linux x86-64: CONFIRMED
- Windows x86-64: CONFIRMED
- macOS ARM64: CONFIRMED (cross-compile via CI)

3. SBOM (Software Bill of Materials)

Embedded in binary manifest JSON:

- All Python packages with exact versions
- SHA3-256 hash of each wheel
- Source repository URL for each dependency
- License (MIT/Apache 2.0 only — no GPL)

4. Verification

```
# Verify binary matches published manifest
python -m anticloud.single_binary verify \
--binary ./wrf_python_binary \
--manifest ./wrf_python_build_manifest.json
# Expected: match=True
```